

CHAPTER 11

ACCIDENTS

DRIVERS WITH GOOD CAR control seldom spin; and if you want to be considered a good racedriver, part of the bargain is to bring the car back to the pit lane with the same number of wheels it had on it when it left. Still, no matter how good you are, accidents happen.

Early in your career, it's likely that you'll go off the road at some point, primarily because you'll be faced with the unexpected without the experience to use the solution that worked "last time." But you try as hard as you can to keep it from happening. Too many new drivers are willing to accept complete losses of control as an acceptable part of driving a racecar, but it is a failure at the most fundamental level.

"Nothing good happens to a racecar when you take it off the track. The faster the car, the worse it is. Keeping the car on the racetrack is part of being a professional."

"Let's put it this way—I've never pushed to the point of the car spinning to find the limit. Guys who think you have to spin to go fast—I guess a lot of those folks have never been to Indy. You wouldn't do it there."

—Bryan Herta

RISK IS INEVITABLE

Fortunately, auto racing today is a much safer sport than it was forty years ago. Today, the cars are generally much stronger, they have real rollover protection, functional driver restraint systems, and, most important of all, fuel containment systems which greatly reduce the risk of fire. Drivers themselves are better protected by helmets, suits, gloves and shoes; and today, racetracks are designed and constructed to reduce the chance of injury.

To be fully prepared for the possibility of accidents, you need to have solutions up your sleeve to minimize the effects of classic racing mistakes.

Still, you can get hurt or killed driving a racecar—any racecar. Even the slowest, most substantially-built car cannot withstand a 90-

m.p.h. impact by another car to the driver's side without injuring the occupant. If you're going to go racing, you have to admit that there are risks. To think that racing has become perfectly safe is a mistake.

Dr. Harlen C. Hunter and Rick Stoff, in their book entitled *Motorsports Medicine*, have done the best research currently available on the risks inherent in racing, and they report the following, based on their most recent research: "It should be safe to assume that the annual incidence of driver fatalities falls between 7 and 14. According to participation and fatality data from the National Safety Council, those rates would put driving race cars in the same hazard range as swimming, alpine ski racing and boating. Race driving would be slightly less risky than scuba diving and mountain hiking and far less dangerous than parachuting and hang-gliding."

"No matter how great I think I was at the height of my career, shit, you still make mistakes. If you're trying, things can [still] go wrong. As much as I or Skip try to tell people everything we know about racing, you ultimately have to learn it for yourself, and part of learning is making mistakes."

—Danny Sullivan

THE SPIN

If your car isn't sliding through the corner, you're not going fast enough. The slide, with a yaw angle that gets the tires to a range of slip angle that maximizes the grip, is your goal. If, however, you over-rotate the car to a point where the momentum of the rear end of

"This is a dangerous business. You know, we put that in the back of our minds and forget it and there's probably not a person in the garage area right now that is saying, 'Oh, this is really dangerous.' It's a business where you know you can get hurt, but you always put that out of your mind. When something bad happens, it's a cold slap in the face and it wakes you up to the fact that it really can be an evil sport and it will bite you. It will bite you in a hurry if you don't treat it with respect."

—Brian Till

the car can't be stopped, and it goes past 90 degrees of yaw, you've spun.

Many new drivers give up too soon and abandon hope on a car that might have been saved. Once a car goes past the point of yaw you wanted for good grip, you have to make a quick mental switch from speed mode to survival mode. When you're in speed mode, you're concentrating on being subtle with yaw angle and control input, trying to balance the car on a fine line that shaves a tenth of a second here and a tenth of a second there from your lap time. When the rear end suddenly pops out 45 degrees more than you intended, style and subtlety be damned—you want gobs of correction *right now*. If one turn of the steering wheel isn't enough, grab another handful and dial in more correction until you've corrected the extreme yaw.

Recovery Is Critical

If you do finally catch it and the velocity of the rotation slows, then stops, at whatever angle, you have to recognize the pause. The pause is your trigger to begin unwinding the opposite lock toward straight, recovering the car. Innumerable spins are caught on the first rotation, but lost again as the car rotates back in the opposite direction with a vengeance, and the steering isn't brought back the other way fast enough.

The recovery has to be slightly ahead of the motion of the car. Slightly behind causes the car to oscillate, the rear sliding out to the right, then left, then right again, then left again—the driver just a little behind each slide. With typical macabre humor, our instructors have developed terms describing these oscillations. The type with big rotations, both right and left, with exhausting corrections by the driver, is called a "tankslapper," a term derived from motorcycle racing, where this type of oscillation literally slaps the tank with the handlebar at the extreme of each rotation. The more subtle, but no less scary version of this problem involves smaller oscillations with higher frequencies and is called the "death wiggle."

In either case, the problem is that the driver is not getting the steering wheel straight fast enough to dis-

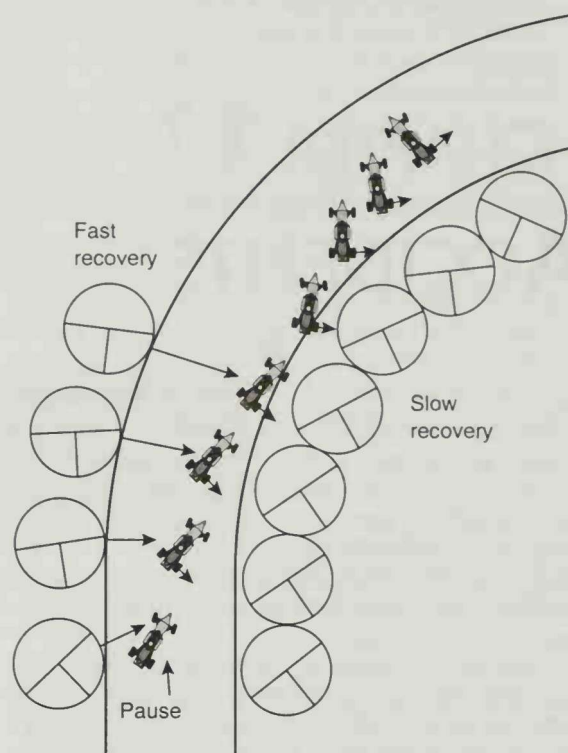


Fig. 11-1. Fast recovery is the key to avoiding second-reaction spins. Once the momentum of the car starts counterclockwise at the pause, the steering wheel has to come back to straight quickly, as on the left. In the steering wheel positions on the right, the steering wheel doesn't come past straight ahead until the spin to the left is beyond saving.

sipate the energy of the car sliding back from its extreme of yaw. This is commonly called over-correction, but *over*-correction is not the problem: any correction that stops the rear from passing the front is *just the right correction*. It's *slow recovery* that causes the problem (see Fig. 11-1).

We've seen tankslappers or death wiggles go on for seven or eight oscillations. The best cure is to just freeze the steering wheel, dead ahead if you can manage it. You won't be correcting for any swerves, but at least you won't be causing any. The car will quickly settle down, perhaps not pointed perfectly down the straightaway, but you can carefully steer it where you want it to go once it settles (see Fig. 11-2).

Spinning's Golden Rule

When it comes to spins, the proven best advice is "If you spin, both feet in."

In any one of these rotations, the first or the sixteenth, if the rotation gets past 90 degrees, the show is over—you've spun, and any further attempt to use throttle or steering will probably make the situation worse. Once it's spun, lock up the brakes and put in

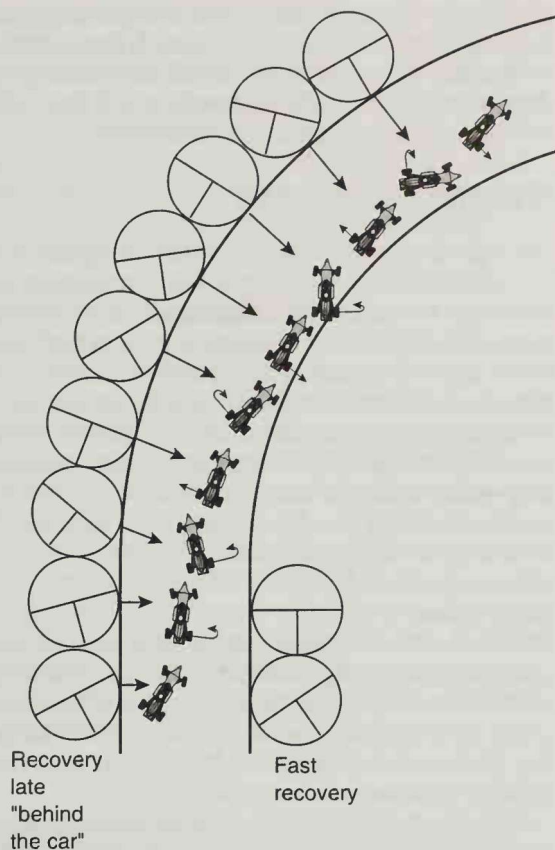


Fig. 11-2. Here there are lots of big corrections, but the recovery has to be quicker to stop the oscillations. Getting the steering wheel straight, quickly, then allowing the car to settle, is the cure for the "tankslapper."

the clutch. If you lock up all four tires they'll lose all their cornering ability and the car will continue on in a straight line tangent to the last arc you were on. This is a real help to others on the racetrack who are trying to avoid hitting your spinning car. If you don't lock the brakes, the car could do one and a half rotations, catch cornering traction, and dart back across the racetrack into the path of other cars—a much more dangerous impact than hitting a tire wall. The only exception to this rule is if, under lock-up, you're heading for a truly substantial stationary object (like a bridge abutment). Here, unlock the brakes and hope that the car changes course toward something softer.

Keeping the brakes locked is also very effective for losing speed, making any subsequent meeting with stationary objects less damaging. With big, sticky slicks on a light car, it's remarkable how quickly they'll stop.

It's key to keep the tires locked up until the car has really stopped. Often, a single car spin is almost over, the car spinning down the middle of the track and all

"Not that I've had a lot of experience along these lines, but locking [the brakes] up is the best thing to do. There is that one in ten where you can release the brakes to keep from hitting something, but that's not something I'd ever teach in a school. Still, if Danny Sullivan hadn't released the brakes at the right point, he wouldn't have won Indy."

—Jeremy Dale

"When I spun at Indy in the 500 that I won, my first thought was, 'Oh shit, I just got into the lead of the Indianapolis 500 and now I'm going to blow it and hit the wall.' I locked up the brakes and dipped the clutch. You're engulfed in smoke, but when it came around the wind was blowing it away from me and it was clear ahead. I had the presence of mind (and it was the only thing I did that was presence of mind) that I took my foot off the brake and gathered it up. The engine was dead from the spin so I had to jump start it and it did another wiggle when I got the clutch out. I was pretty lucky."

—Danny Sullivan

the traffic finds a clear way by, and suddenly the driver in the spinning car releases the brake at 5 m.p.h. and rolls in front of a following car that would have made it by if the hole didn't close. In a spin, lock them up and keep them locked until the car *absolutely stops*.

If you successfully push in the clutch when the spin starts and heel and toe to keep the engine running, you can go back to first gear (even in mid-spin) and be ready to get out of the way of following cars. If you can't see oncoming traffic clearly, stay put. Look to the nearest corner station and they'll help you re-enter safely. The accepted method is for a worker to point repeatedly at the car they want you to re-enter behind, then wave you on as it goes by. Try not to be so impatient and frustrated with the spin that you spin the tires on re-entry and spin again, unless, of course, you like to provide extra amusement for the fans.

Sometimes the spin doesn't have a happy ending. If you get off the racetrack and make it to a barrier, your race is over. If the car has hit and stopped, you'll be surprised to find that your first instinct is often to try to back it out of the barrier and get back on the track. Your competitive blood is still up and you find yourself thinking, "It can't be *that* much slower with three wheels." When you're tempted, just give a moment's thought to the additional damage you can do to the car and yourself by trying to drive a twisted car

back to the pits. Then again, in 1990, Barber Saab champion-to-be Rob Wilson drove the final 3 laps of the Tampa round on three wheels to finish 9th out of a field of 18 cars. Some guys just never give up.

HITTING HARD

If you hit hard and it's obvious that you're not going anywhere, the first thing you should do is to shut off the master electrical switch, reducing the chance that an errant spark could set off an oil or gasoline fire.

There are actually things you can do to help protect yourself between the time you lose it and the time you hit. Your first instinct, especially in a Formula car, is to scrunch down and hide in the cockpit, squeezing the stuffing out of the steering wheel. You can reduce the chance of injury by doing the exact opposite: if you hold onto the wheel tightly and the impact involves the front wheels, the steering wheel may spin violently, damaging your hands or wrists. It's against your nature to let go of the wheel, but if impact is inevitable, it's the best thing to do. If it's going to be a frontal impact, you might want to lean forward and put your chin on your chest. It probably won't help to take your feet back from the pedals since the Gs of impact will be stronger than your efforts to keep them back. In a rear impact, you want to push your back against the seat back and your helmet against the headrest to lessen the chance of whiplash.

If it's never happened to you, it's hard to believe that you have time to think about these things, but as a rule, most drivers feel like time passes very slowly between the time they lose control and the time they hit something.

Stay in the Car

If you hit really hard, once you've turned off the master switch, resist the temptation to undo the belts and jump right out. The only reason to hustle out of the car is if it is burning, and believe me, you won't need our advice to get out if it is. If it's not burning, take some time to assess the situation. If you crashed because of a sudden downpour or oil on the course, more cars may be along in a moment and you certainly don't want to be half in and half out of the car when another one runs into it. You're safest securely belted inside a substantial roll cage in these situations.

Even if there's little chance of another car coming along and joining in on your accident, stay put for a minute and assess whether you're okay. If you're dizzy, have back pain, neck pain or numbness anywhere, don't move or thrash about. You may have an injury that you can aggravate by moving. Don't take your helmet off. Sit still and wait for the emergency workers to arrive. They are trained in emergency procedures, extrication techniques, and fire fighting, and they are your best chance of getting through in the best shape possible.

If your self-assessment reveals that you're physically okay (you'll certainly be mentally depressed), look back and make sure that traffic isn't coming your way before you undo the belts and get out. The corner workers will direct you to the safest place.

COMMON ERRORS

Once you're physically okay, the first thing you should do while the memory is fresh, and before you have time to make up rationalizations, is to try to figure out why the accident happened. Spins and crashes are most often caused by mistakes and not equipment failures. Unless you'd like to make the same mistake again in the future, with the same result, you'll want to identify the error and cure it. In order to help define what you did wrong, you can use the following shopping list of common errors, some of which we have covered earlier.

Early Apex

If the spin came in the second part of the corner, especially from dropping wheels off at the corner exit, this classic line mistake can be at fault. Sometime when you're on the track, take a look at the skid marks coming off of corners (see Fig. 11-3). They tell the sad story of how common this type of spin is.

Sensing that the car is running out of road can lead you to turning the steering wheel more, pinching off the cornering radius late in the turn. If the car isn't already at the limit, it will get closer to it or over it if you tighten the arc. If the car is already sliding and drops wheels into the loose stuff at the exit, it will rotate fast.



Fig. 11-3. Notice the path of the skidmarks here at the exit of Turn 7 at Sebring, all caused by early apexes.

Jerking the Car Back On

Sometimes the car drops the two outside wheels off the track coming off the corner, most frequently as a result of an early apex; and the inexperienced driver tends to act too suddenly, jerking the car back onto the racetrack. If you use a lot of steering lock very sud-

denly, the outside front tire which starts out in the dirt doesn't provide much direction change—at first. When it makes it from the dirt back to the asphalt, it takes a big bite and hurls the front of the car violently toward the inside of the course. The back end of the car, of course, comes around too quickly for you to save. The accident looks like it's going to happen on the outside, but the car ends up on the opposite side of the track, often unnecessarily (see Fig. 11-4).

You can ride out this mistake by straightening the steering wheel and reducing the cornering demands on the car to zero. With the wheel straight, the car can bump along in a straight line, two wheels on and two wheels off, more or less under control. Next, you gently coax the car back onto the racetrack with tiny steering wheel inputs over 100 to 150 yards or so.

TTO

Trailing throttle oversteer (TTO) will sometimes get the tail out so suddenly that it can't be saved, especially if the throttle is snapped off, not relaxed. It can happen anywhere in the corner, from turn-in to track-out. To identify it, you have to think clearly about what your right foot is doing on the throttle throughout the course of the corner.

Often TTO is combined with other mistakes, like the early apex, to cause the spin.

Mismatched Downshift

In most cases, you'll choose to do a downshift on the straight approaching the corner, but there are plenty of situations where you're forced to do it while turning. Going down a gear while cornering and doing it badly without matching the engine speed to wheel speed can spin a car instantly, almost as fast as yanking up on the handbrake.

Going Too Deep

It's easy, especially when you're running at a familiar racetrack, to get overly optimistic about your brake points and wait too long before going to the brake pedal, arriving at the turn-in point too fast. If the corner has an escape road, you can just go straight on and give up the corner. Without runoff, you try to turn down into the corner, but remember that the radius of the arc is a direct function of the speed. With decent car control, you won't necessarily spin as long as the car stays on the asphalt. The only way to make the apex is to be close to the speed that coincides with an arc that reaches the apex. Too fast, and you're likely to



Fig. 11-4. Here at the trackout of Turn 2 at Sebring, the driver dropped two wheels off at the exit, then hurriedly tried to force the car back onto the racetrack. When the outside front wheel came off the sand, back onto the asphalt, the car turned sharply left.

miss the apex because the arc you're committed to is too broad to make it down to the inside of the corner. How far off you are from the apex is a function of how overly fast you're going. Slightly too fast, and you miss it by a couple of feet. Way too fast, and you might miss it by the width of a racetrack. Once you're on the grass, the party's over.

If you routinely use the braking and turning ability of the car, getting in too fast can be saved by continuing to slow down while bending the car into the corner. Trail the brakes as deep into the corner as it takes to slow it down to a speed that keeps it off the grass. If your habit is to do only straight-line braking, the possible fixes to arriving at the turn-in point too fast are limited to continuing to brake straight until you run out of road. Braking and turning makes a lot more of the road and many more options available to you.

"To me, one of the most valuable things about braking and turning is saving your butt when you take too big a bite in braking. Most corners, you don't brake and turn way down to the apex—it's more often a trail brake—so if you do get to the turn-in point going too fast, you just extend that zone down deep into the corner, bowing the car toward the apex. It's a valuable asset."

—Terry Earwood

Failing to Warm Up

A similar situation occurs when you fail to get the brakes or tires up to operating temperature and try to go to the normal brake point with a car that isn't prepared to perform at its usual level of braking. If the brake pads and rotors are cold, they don't create the same braking friction that they do when hot. If you step on the brake pedal with the usual force at the usual place and the brakes are working at only half of their capability, the result is that you arrive at the corner entry going way too fast.

Even if the braking system is up to temperature, a fresh, cold set of tires don't generate their grip until they warm up. Slicks, for example, operate best at temperatures over 200 degrees Fahrenheit. Roar out of the pits with 65-degree tire temperatures and the tires will lock up if you push on the brake pedal with your accustomed level of pressure. Push more lightly to avoid lock-up, and it takes longer to slow down. To fix this problem, warm up the brakes and tires before flirting with your normal brake points.

If you haven't been in a racecar for a while, you can lose the sensitivity to just how hard you can push on the brake pedal in a racing situation as compared to the street-driving world. Often, cars get into a corner

too fast not because the driver went too deep but because he or she went to a brake point that would have worked if all of the car's braking potential was being used. Every time out, you have to re-establish the level of brake pedal force for that day. You do so by intentionally braking earlier than the traditional brake point and pushing harder and harder, searching for a trace of lock-up. Once you find the threshold, you can then work the brake points closer to the corners.

Stackup

Corner entry can be tricky enough when running alone, but stirring other drivers and cars into the mix can bring on additional complications. One common cause of problems is not anticipating the brake application of the car ahead and, as a result, either drilling the lead car from behind or losing control in the swerve to avoid it. You have to make a braking adjustment when following another car closely. You can't afford to focus on your traditional brake point and be blindly determined to go to that spot before you go to the brake. This holds especially true if you're gaining on a slower driver. If you're catching the car ahead, it's likely that the other driver is losing some time to you in the braking zone, for one reason or another, braking early. You've got to anticipate this and adjust your brake point based on what you've seen the other driver do in the last few laps.

Multiple car trains are another prime opportunity for a stackup. If six or seven cars are in line, even though they are turning exactly the same lap times, the last car in line has to brake when the first one does. If each car is 15 feet long, the last car in a seven-car train has to put the brakes on 105 feet sooner than the first car in order to avoid running into the car immediately ahead (see Fig. 11-5). This situation arises most often at the start of a race when the cars are the most closely bunched, and it's a key problem to remember before each race start.



Fig. 11-5. The driver of car No. 19 didn't anticipate that coming into the Hairpin sixth in line required a much earlier brake point than normal. Luckily, the car contact with the rear of car No. 24 resulted in minimal damage to both cars.

Passing

Many car-to-car incidents take place in passing situations, as you saw in the earlier chapter on passing. For example, waiting until the last possible instant to pass, then jerking the car suddenly out of the wake of the car ahead, is the cause of many an accident. The violence of the swerve can upset the car so badly, it can spin on the straight. Another dangerous move is get the pass wrong and clip the other car as you go by: this move leads to air time in open-wheel cars. Pulling back in line too soon is another common boner, easily solved by taking a quick glance in the mirror.

Sometimes the driver of the passing car figures that two objects should be able to occupy the same space at the same time, forgetting that it is the responsibility of the overtaking car to pull off the pass without contact. The net result is most often two cars eliminated from the race. The sad thing in this type of situation is that someone else's error can take you out: sometimes, you're the victim.

BEING THE VICTIM

All of the previous categories have been incidents that resulted from mistakes. But the mere act of putting yourself in a racecar exposes you to the possibility of accident through no fault of your own. To quote the classic line in motorsports, "That's Racing."

Sudden Oil

If you're the first to arrive at a corner at which the previous driver coated the line with oil from an exploded motor, you're probably going to spin and the severity of the accident will be a matter of luck. The same is true for other common racetrack lubricants like engine coolant, sudden rain or sand kicked up on the track from a previous spin. The corner workers will try to warn you with the appropriate flag, but if you're the first one along, they might not be in time.

Oil and coolant leave a trail on the racetrack if the motor failure is the slow, agonizing type rather than the sudden *boom* type. With both oil and coolant, you can see the sheen they give to the racetrack, especially if the sun is low in the sky. If the track suddenly looks shiny, try braking early to see what you've got, reserving the option to head off-line to get some braking grip back if your suspicions are confirmed. Heading off-line, or at least off the oil path, is the best option in these situations, since an oiled racetrack is sometimes difficult to stand up on, let alone drive on.

Breakages

Sometimes the stupid cretin that's oiling down the racetrack is you. If the motor is going away and oil and coolant are spraying out the back, this glop can get directly on your tires and spin you—an event all the other drivers will applaud. If it happens suddenly, as in the motor going off like a grenade, it's not your

"I've been surprised by an oiled-down corner more times than I can count. It's easy to think, 'Oh well, I'm going to crash,' but if you keep your wits and don't give up, you can save it. You just have to switch over to survival mode and forget about speed, forget about line and scrape through the corner, even if you have to nearly stop. The oil's not everywhere, so once you spot it, you get off onto a part of the racetrack that's dry. You learn not to correct the steering wheel too much or to do anything sudden or violent while you're in the oil—let the car slide. If you do a big correction, then hit the good part of the track again, the car will hook and off you go. Just stick with it and don't quit. Like pilots say, 'Fly it all the way to the ground.'"

—David Loring

"As a group, we racers are so optimistic about how well things are going to turn out each day that we turn our backs on all the things that could go wrong.... There's a certain amount of denial at work. A piece on the car could fail at the kink at Road America or any of a dozen places that wouldn't be good. A car's a machine and machines sometimes break. Happens to the best of them. Rick Mears, driving for one of the best teams in the world—a radiator hose blows at Indy and puts him in the wall.

"You have to be concerned before you get in the car. Is the car I'm getting in safe? Are the guys who are preparing the car top notch? Is the car owner supplying the proper parts or are we pushing it? I remember it was a big day when I had the opportunity to drive a car and I said no. Thank you, but no thank you. You don't have to drive every one that you can."

—Jim Pace

fault (whatever consolation that may be), but if the water temperature is rising and the oil pressure is falling, you can't see the cars behind you through the smoke, and yet you continue on, you're being a fool. A totally exploded motor is much more expensive to repair or replace than one that still has most of its parts bolted together, no matter how worn or hot those parts may be. If the motor is on its way out, get off-line—make the effort to avoid ruining the race for everyone else.

Any component—a motor, a tire, some key part of the braking system, a rod end, an upright, a wheel—

can fail unpredictably, with disastrous results. No amount of money or engineering expertise can overcome this possibility. As a driver, you have to accept this fact as part of the game—that through no fault of your own, the car can betray you and hurt you. You don't dwell on it and you certainly don't let it affect your desire to take the car to its limits—it's just part of the deal. The best defense is to make sure that you drive well-prepared racecars and that you own and use good safety equipment.

PREPARE FOR THE WORST

The best advice in preparing for the possibility of an accident is to outfit yourself and your racecar with the best safety equipment you can acquire.

The Car

The car has to have a six-point belt with a submarine strap, attached to failure-proof mounts. The rollover bar or roll cage should be stout and plenty high enough to keep your head off the racetrack in case of a rollover. The major race-sanctioning bodies have stringent and well-proven guidelines on the level of car preparation required to run.

You need to be careful and consider the risk involved in competing in club events in street cars unequipped with roll cages or fuel cells, and with no more than a stock seatbelt anchor. These cars, even driven singly against a clock, can get upside down. We won't try to dissuade drivers from competing in this branch of the sport, but they should be aware that in many ways, these cars are more dangerous than a fully-prepared racecar.

Driving Gear

Your driving gear should include the best full-face helmet you can afford (the old adage is that you should get a \$50 helmet if you have a \$50 head). Open-face helmets are common in full-bodied racecars, but the full-face helmet is no more uncomfortable than an open-face and offers more protection to the side of your head people see most (see Fig. 11-6).

Your fire resistant suit should be at least two layers thick; and gloves, socks, racing shoes, and a hood (if you have facial hair) should be worn whenever you're in a racecar (see Fig. 11-7).

Neck collars are becoming increasingly popular and it would be valuable to try one out (see Fig. 11-8). Some drivers find them restrictive, while others find the support they give to the neck and helmet valuable.

In open cars, arm restraints (see Fig. 11-9), which give you freedom of movement but keep your arms from flying out of the cockpit in an upset, are required in some forms of racing but not others. It's hard to come up with an argument for not using all the safety gear you can find—it's normally not a hard sell.



Fig. 11-6. Buy the best helmet you can afford, with the current Snell test standard.

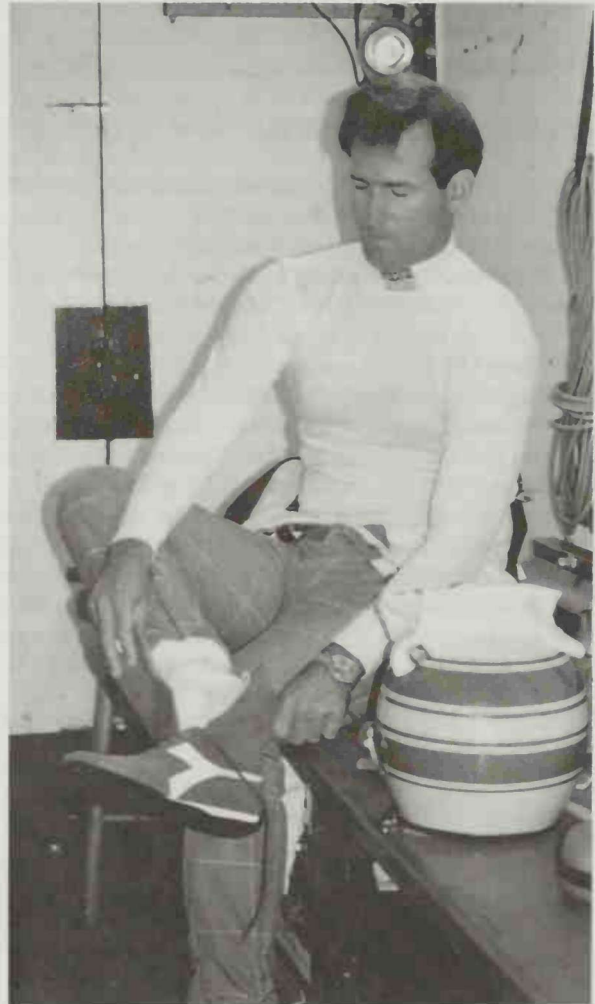


Fig. 11-7. Fire-resistant apparel and proper racing attire are crucial to safety.



Fig. 11-8. Neck collars that restrict helmet motion under deceleration are increasingly popular.



Fig. 11-9. Some sanctioning groups require arm restraints in open cars to limit the travel of your arms outside the cockpit.

Sometimes safety gear is not used properly, reducing its effectiveness. The most common safety mistake is to not do up your belts tightly enough. You should start with getting the lap belt tight—really tight—low down across your hips, not up high on the soft part of your abdomen (see Fig. 11-10).

Next, the shoulder belts can be tightened. It's important to do it the other way around—shoulders first, lap second—because the lap belt rides up too high into your belly area. The submarine belts are the only ones that don't need to be ultra-tight, for comfort's sake and because their major job is to keep you from sliding out from under the other belts.

As you or a helper tighten the belts, wiggle around, trying to take as much slack out of the belt system as you can. Once you get out on the racetrack, the cornering and braking loads will push you around in the seat, loosening the belts. They're hard to adjust underway, so you really have to get them right before you leave the pit lane, perhaps having them uncomfort-



Fig. 11-10. Tightening the belts, really tight, starts with the lap belts first. Use a helper if need be.



Fig. 11-11. Always check the security of the helmet chin strap, then check again before leaving the pit lane.

ably tight in anticipation that a few laps will loosen them. There's a performance advantage in being securely belted into the car. If there is no need to brace yourself to keep from sliding around the cockpit, you'll find that your muscle tension will be lower and

you'll be able to make more accurate motions on the controls. Any extraneous motion side to side and fore and aft has to make it more difficult for your central nervous system to sense the subtle accelerations involved with car control. So belt up tight.

Silly as it sounds, drivers sometimes forget to do up the chinstraps on their helmets. The helmet won't do you any good if it doesn't stay on your head. Check the chin strap every time (see Fig. 11-11). It's a good habit when leaving the pit lane to tug on the shoulder belts one more time and to check the helmet strap.

LAUGH AND THE WORLD LAUGHS WITH YOU

Drivers have developed a version of gallows humor about accidents—partly from sheer amazement that they got through it, partly to deflect the dead seriousness of the subject. Here are just a few examples.

Soon after a huge accident that he had in Florida, Mike Rand, former Formula C National Champion, was recounting to Price Cobb how it all happened. "I got down to the end of the straight and Shelton, who had just drafted me, pulled back too soon and hit my left front and the car went *bang!* straight up in the air. I closed my eyes—"

"Closed your eyes?" interrupted Price, "You missed the best part."

Former World Champion Jody Scheckter describes a big crash: "I slid down in the seat as far as I could, closed my eyes and waited for all the banging

to stop. Then it gets quiet, and that's the tricky bit. You don't want to undo the belts quite yet because the quiet part means either it's over or you're flying through the air and the banging may start up again."

Dorsey Schroeder had the opportunity to compete in the International Race of Champions and got a rare chance to race at Talladega, one of the fastest tri-ovals in the world: "We were running constantly flat out, never even thought about lifting, and I was doing well, running 6th when I got tapped. I must have been going 185 m.p.h. and it came around and there was nothing I could do about it. I had never spun at that speed before and I thought, well, this is a new experience. I had seen Rusty and Dale and other people go on big rides and I thought, 'so this is what it's like.'"

"I didn't like it. Then the car takes off into the air, still going a zillion miles per hour and I'm thinking 'Oh, no.' It got down on the infield grass, which was soaking wet, upside down, and it's just scooping mud and water and I don't know what else, all inside the car into my helmet, in my eyes. It was terrible.

"The car finally stopped and I thought, 'Now that I know what it's like, I definitely don't want to do that ever again.' I undid the belts and crawled out of the window, absolutely covered from head to toe with mud, dripping wet. The ambulance got there a minute later and roared up to me. The back doors to the ambulance flew open and the girl EMT inside took one look at me and said, 'You're not getting in *my* ambulance looking like *that*.'"

There's a million of 'em.